



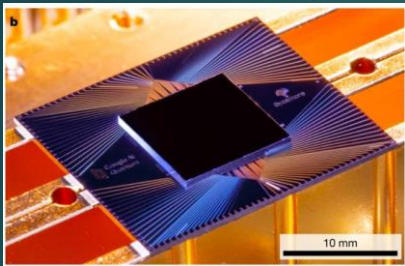
An Empirical Pilot Study of Training Dynamics in Hybrid Quantum–Classical Neural Networks on Real-World Data

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Background and context

Hardware advances

- ▶ 2019: Google sycamore



- ▶ 2024: Google Willow



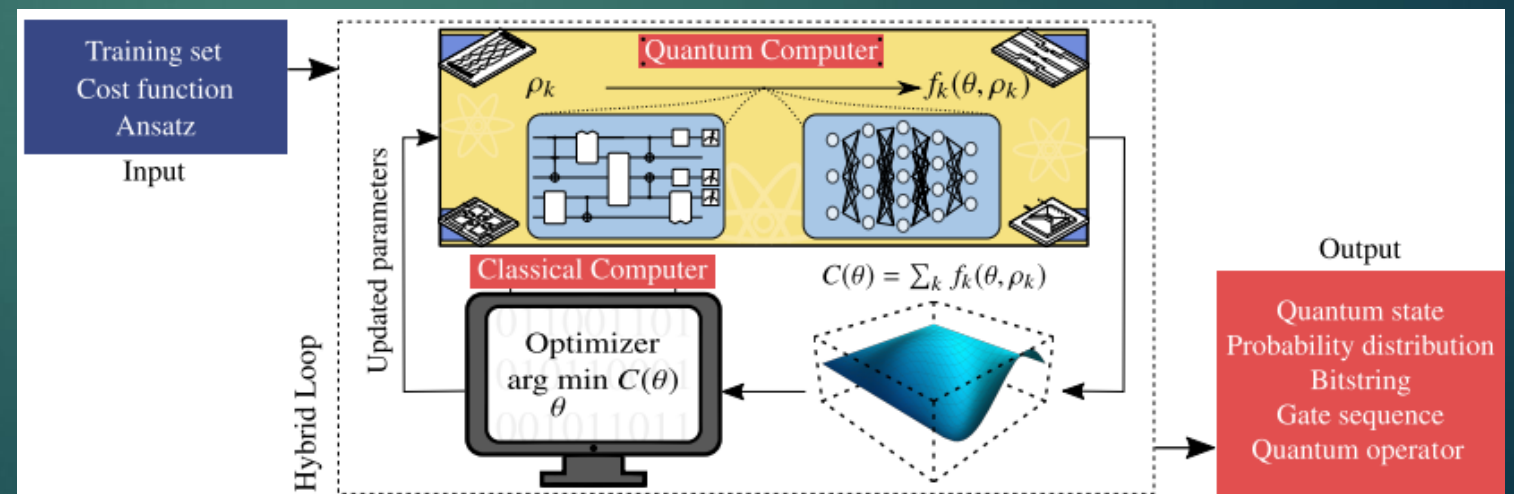
- ▶ 2025: Google quantum echo

Quantum In finance and Industry interest

- ▶ Applied to Pricing, Risk calculation and trading tasks
- ▶ IEEE QCE 2024 had a dedicated talk on “Quantum algorithms for financial applications”

QML, VQA and QNNs

- ▶ Larger expressivity via features of quantum + classical training methods



Training dynamics and obstacles in training hybrid quantum-classical models

1) Barren Plateaus

- quantum loss landscape becomes exponentially flat, making gradient-based training increasingly difficult as the system scales. [3,4]

- where

- $$\text{Var} \left[\frac{\partial C}{\partial \theta_i} \right] \sim O(e^{-n}) \quad \& \quad C(\theta) = \langle \psi_0 | U^\dagger(\theta) H U(\theta) | \psi_0 \rangle$$

- It is characterized by the variance of gradient becomes exponentially small. (similar to vanishing gradient)
- Types: Expressivity induced, Entanglement induced, Noise and Data induced [4]

2) Architecture dependence

- Even in the absence of barren plateaus, poor architectural choices can render a model untrainable. [4,6,7]

3) Data dependence

- Performance depends heavily on structure and encoding of data into quantum state [8, 21]

4) Ansatz dependence

- The choice of parameterized quantum circuit (ansatz) affects both expressivity and trainability. [5,6]

Types: Problem specific and Hardware efficient

BUT! How do you know if your model is good? Or how much is the quantum layer helping?

Why we need a framework

Majority of current studies rely on isolated metrics and synthetic data [2] resulting in bias towards positive results and false optimism about the ‘learning’ of the models, with less focus towards real-world application and fair benchmarking.

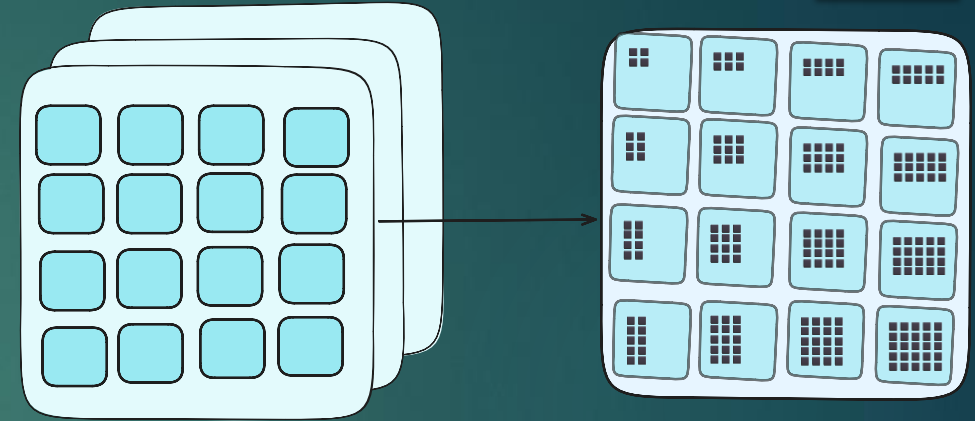
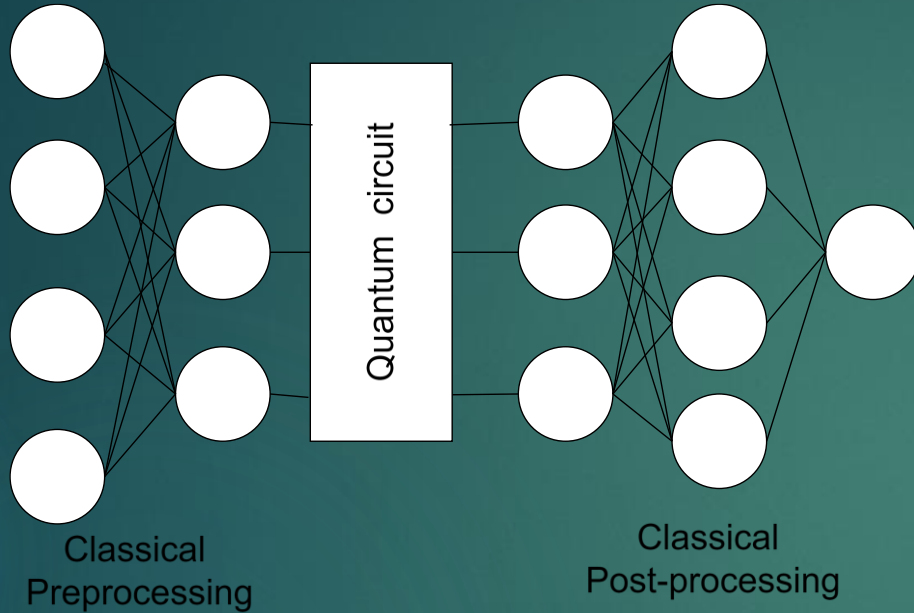
But! Existing frameworks are narrow* so, I propose the use of following metrics to evaluate different aspects of a hybrid quantum-classical neural network as:

- (1) Convergence (loss): Does the model as a whole converge for the given cost function?
- (2) Trainability (gradient norm of quantum layer): Does the quantum layer in particular learn anything?
- (3) Utility (quantum contribution score): How does the addition of the quantum layer reflect in real-world performance?

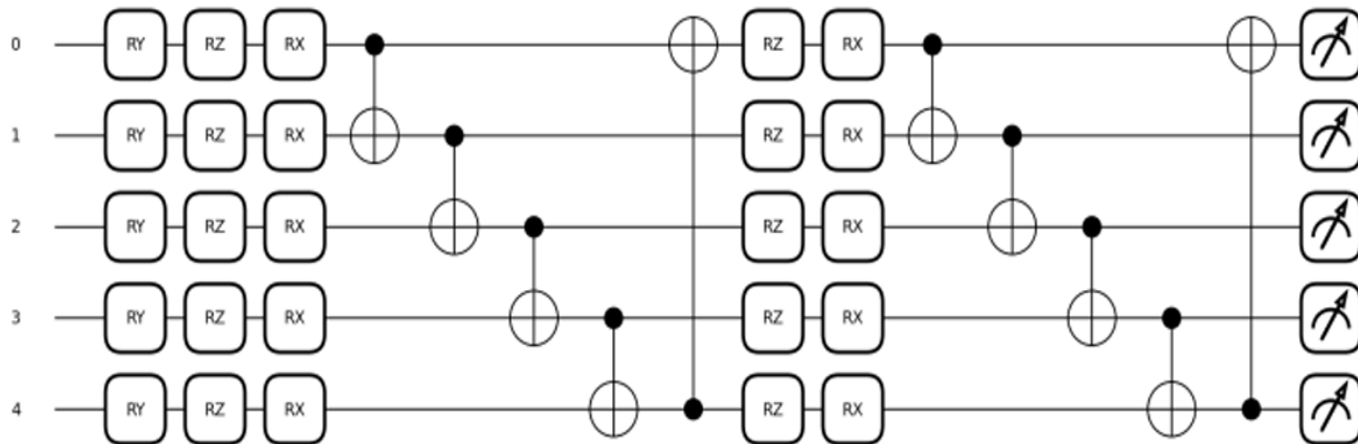
- ▶ * [2] have a benchmarking framework but it is narrow and focuses on classification tasks on synthetic data
- ▶ [2] Bowles et al. "Better than classical? The subtle art of benchmarking quantum machine learning models." arXiv:2403.07059 (2024)
- ▶ [3] McClean, J.R. et al. "Barren plateaus in quantum neural network training landscapes." Nat Commun 9, 4812 (2018).

Experimental setup, architecture and Ansatz

Quantum circuit: VQA (HEA) and hybrid architecture



16 representative models averaged across
3 random seeds Spanning 2-5 qubits and
2-5 layers



- Data:

Real-world: NIFTY 50 option pricing data (2014–2018), preprocessed as per "Approach 3" in Goswami et al. [19]
Synthetic: Generated using `make_regression` to resemble real-world data in terms of noise and redundancy

Observations

1. Loss and gradient disagreement. (Convergence $\neq$ trainability)
2. Gradient suppression and resurgence and correlation with model architecture.
3. Quantum Utility and Trends in QCS (difference in data, relation with convergence and trainability)

Key take-away: Convergence $\neq$ Trainability $\neq$ Utility

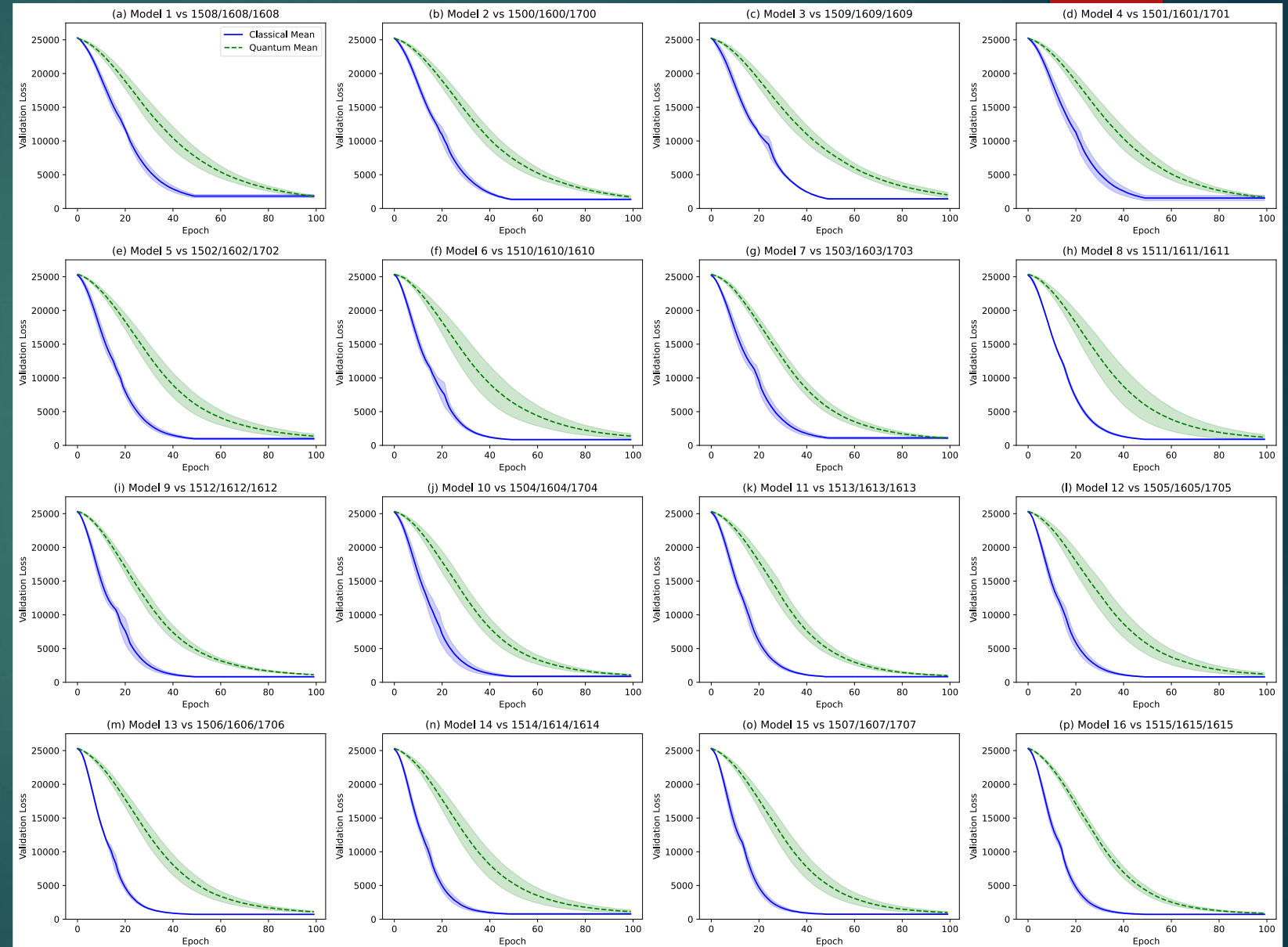
Additional: Gradient in real and synthetic datasets

1) Disagreement between loss and gradient

Loss curves for all models converge

Key observation:
Convergence $\neq$ Trainability

- How does it relate to literature?
 - Classical Compensation Hypothesis: [26] Wang et al. showed that classical layers can "mask" quantum layer difficulties by providing a stable baseline
 - Limitation of single metric: [2] Bowles et al. emphasized fair benchmarking requires multiple metrics



2) Gradient norm analysis

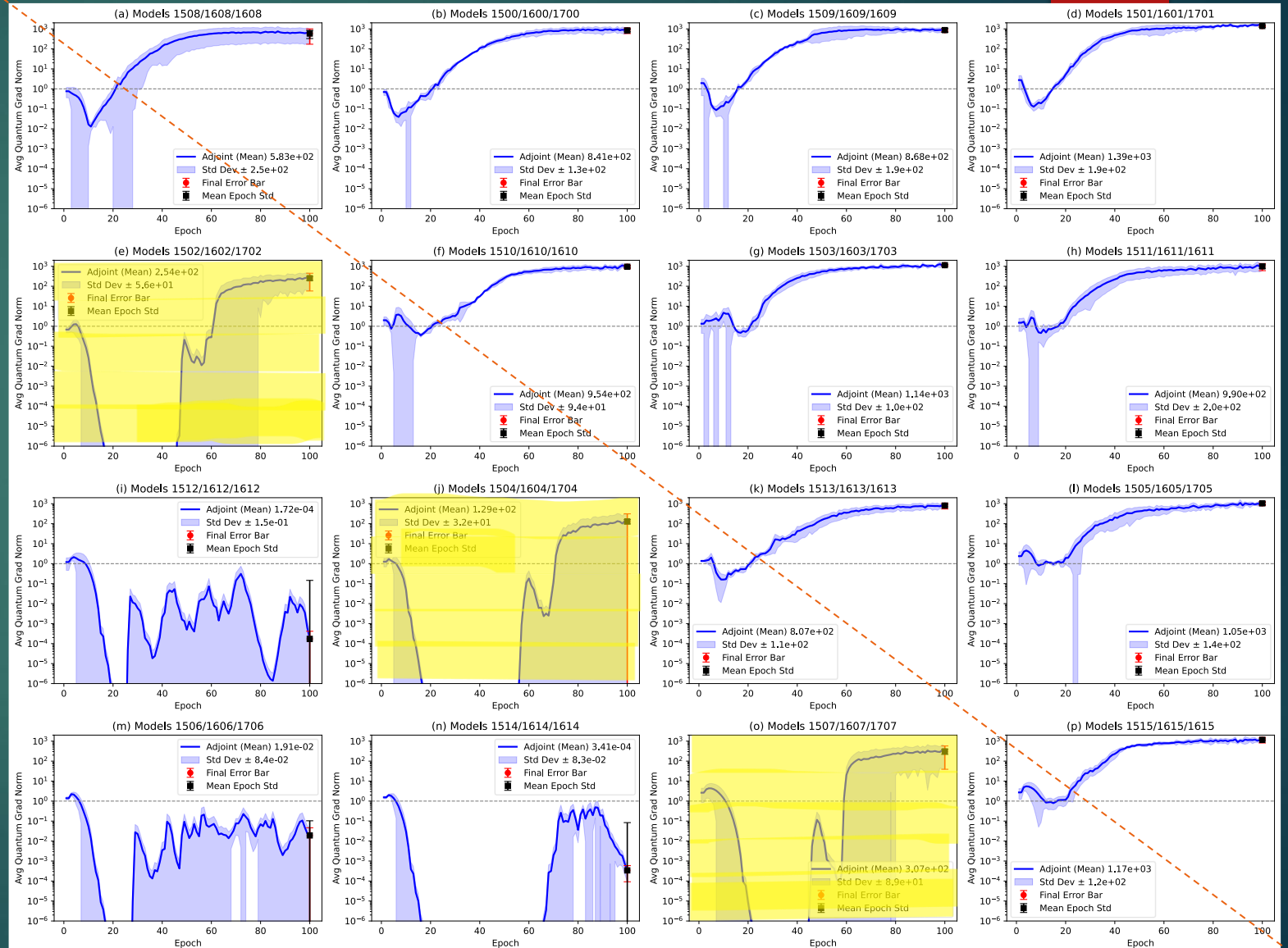
► What do I observe ?

- a) Gradient is less stable and suppressed for wider circuits, consistent with [1,3,4]
- b) Correlates with Q/L ratio with symmetry along the dotted orange line.
- c) In some models gradient re-surfaces at late epochs after being suppressed early on [26]

► Why could this happen (hypothesis), implications?

- Entanglement-Expressivity Balance: [22]

- Classical layers changing over time could dynamically alter the quantum optimization landscape. [26]



3) Quantum Contribution Score

$$\text{QCS} = 1 - \frac{\text{HQNN_validation_loss}}{\text{Classical_ANN_validation_loss}}$$

- ▶ QCS = 0 (no advantage), QCS > 0 (quantum layer helps), QCS < 0 (quantum layer hurts)
- ▶ Why did I choose it too be as it is: corresponding classical baseline, validation loss as performance proxy (reflects real-world generalization on unseen data), unbounded –ve value to focus more when the quantum layer hurts rather than helps.

Trainability ≠ utility

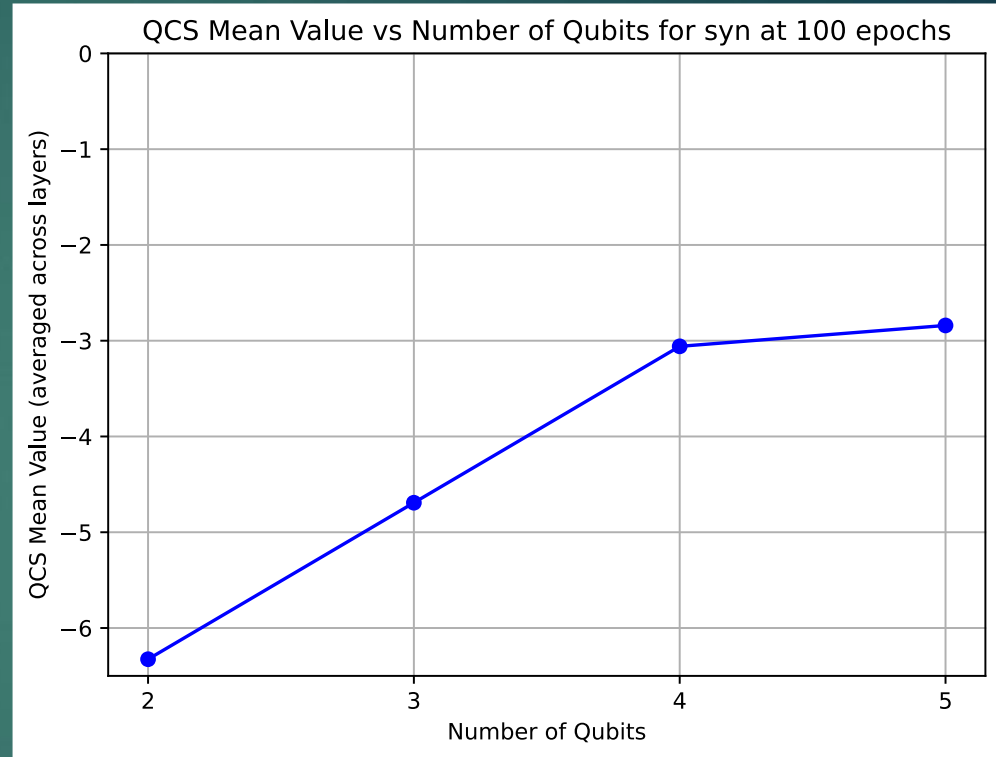
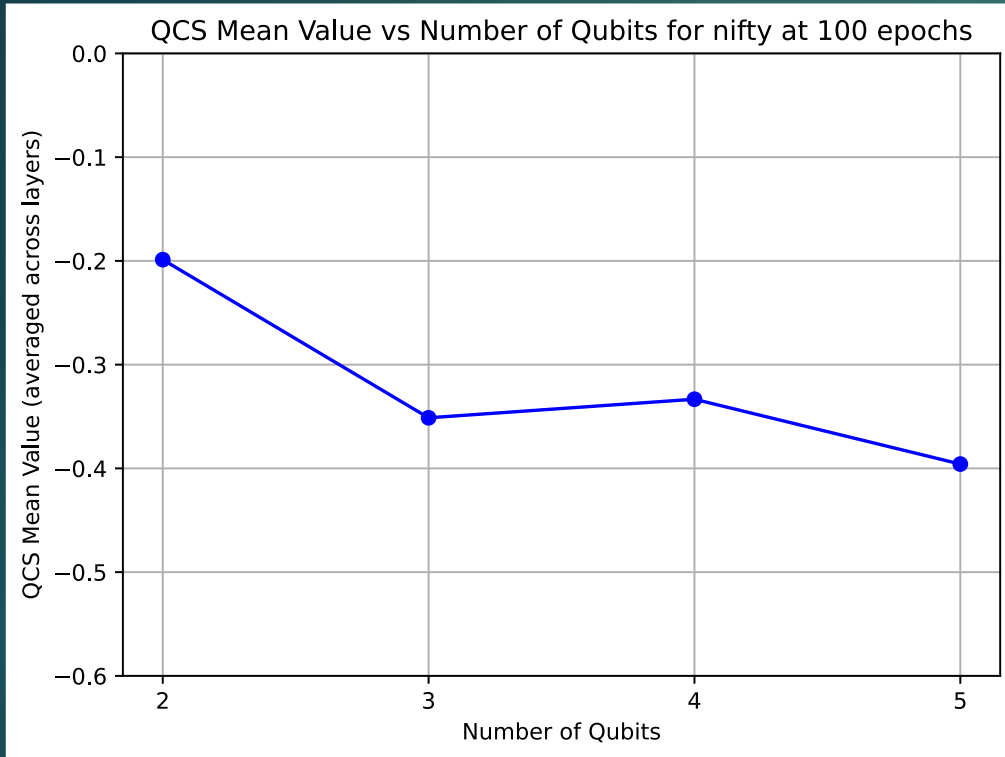
Possibly better approaches to quantify utility ?

- Taking into consideration the additional cost of parameters, or a well behaved formulation with balanced read instead of loss focused.

[2] Bowles et al. (2024) Benchmarking Framework: a parameter matched framework to compare quantum and classical models for classification but not a numerical measure that isolates the quantum layers performance

3a) Trends in QCS with architecture

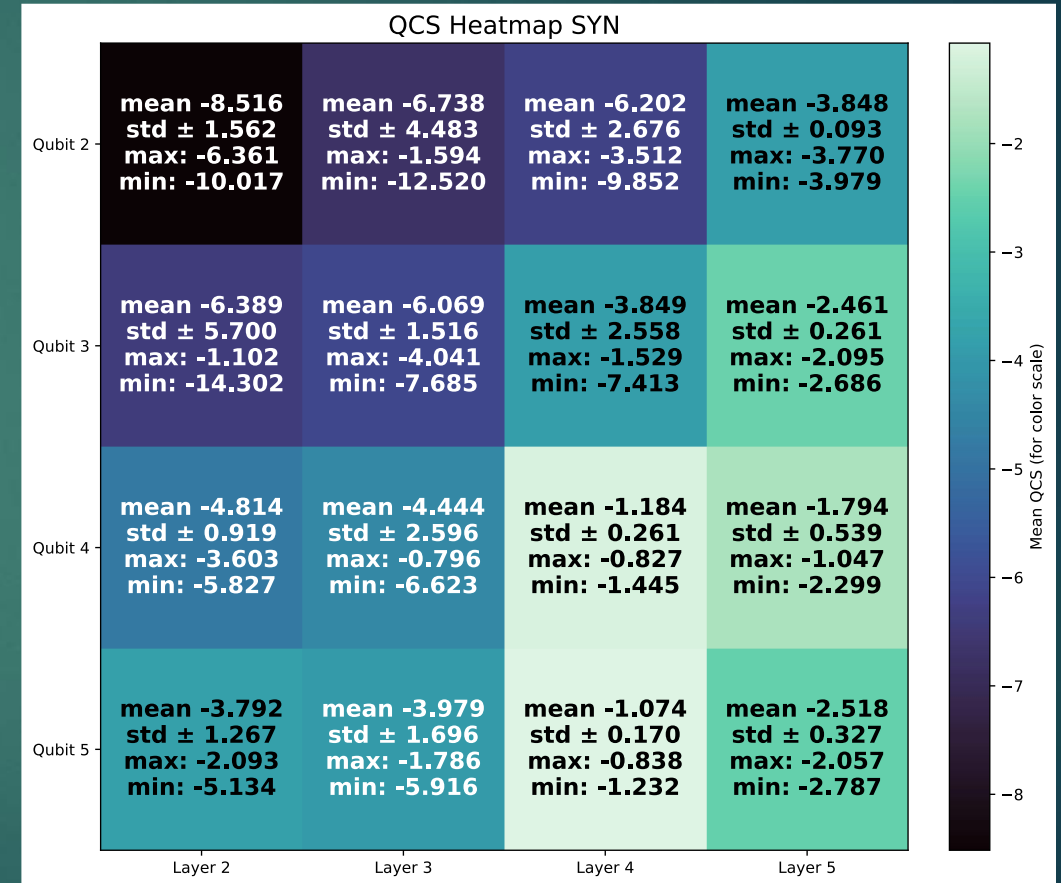
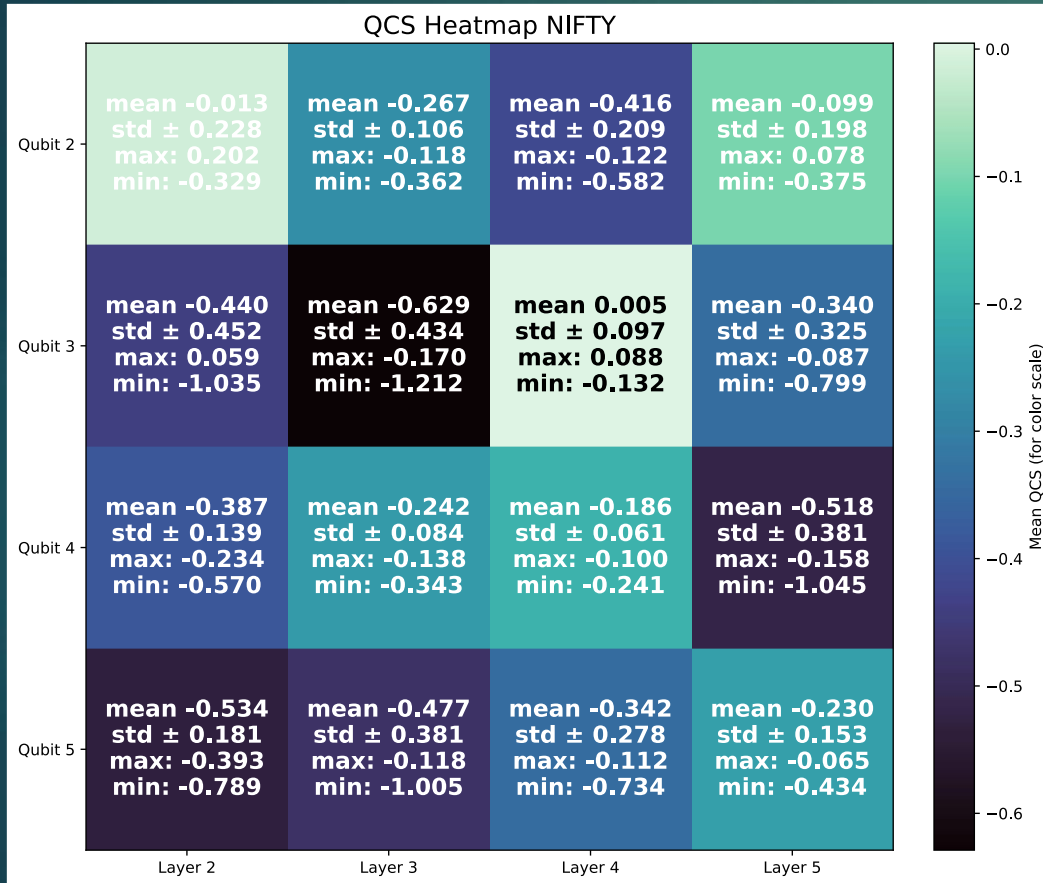
- Strong correlation was observed between QCS and number of qubits for synthetic data, but not for real data



- The Power of Data in Quantum Machine Learning [8]: The "power of data" determines whether quantum models can outperform classical ones
- Data Encoding and Expressive Power [21]
- Hypothesis: Data lacks exploitable structure, Encoding was sub-optimal and ineffective, Classical models were already optimal

3b) Trends in QCS with data

- No particularly strong trend for real-world data, but uniform trend along both depth and width for synthetic data.
- False Optimism from synthetic data (in terms of trend in utility) BUT! Better (less negative) QCS score for real-world data



- Note: the scales are not matched for both.

Limitations

- ▶ Limited ablation studies (extend across circuit type, seeds, variables(learning rates for the gradient observations))
- ▶ How to design a quantum circuit to outperform classical counterparts, than naively guessing

Future work

- ▶ Investigate if the resurgence of gradients can explain more about quantum-classical interaction and
- ▶ Evaluating data characteristics/structure and other aspects of the model like amount of entanglement (as studies have shown entanglement as a contributor to performance/BP)
- ▶ Consider extra costs in training quantum circuit while evaluating utility
- ▶ More seeds for statistical significance
- ▶ Test this in other domains/tasks/data/ansatz/larger ablation studies

Conclusion & Summary

- ▶ a multi-axis diagnostic framework (Loss + Gradient Norm + QCS)
- ▶ to evaluate hybrid quantum-classical neural networks.
- ▶ The framework reveals that Convergence $\neq$ Trainability $\neq$ Utility,
- ▶ highlighting that single metrics are insufficient to diagnose HQNNs.
- ▶ Loss tells you the model is training. Gradients tell you the quantum layer is learning.
- ▶ QCS tells you the quantum layer adds value. You need all three.
- ▶ Synthetic data is misleading—test on real data.

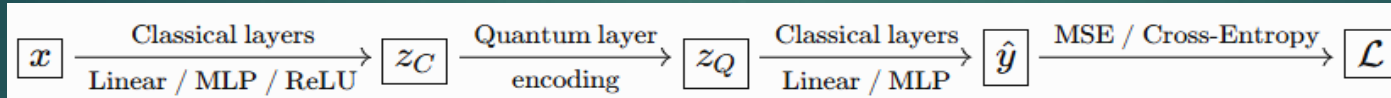
Thank you

Hybrid quantum-classical models

1) Why bother with classical layers at all?

2) Training of Hybrid quantum-classical models [24]

- Forward pass:



- Backward pass:

$$L \longrightarrow \nabla_{\theta_C} L, \nabla_{\theta_Q} L \longrightarrow \theta_C, \theta_Q \text{ updated}$$

Where $\nabla_{\theta} L = \left[\frac{\partial L}{\partial \theta_C}, \frac{\partial L}{\partial \theta_Q} \right]$ and parameter update rule is $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L_t$

Quantum Layer

$$|\psi(x)\rangle = U_{\text{enc}}(x)|0\rangle^{\otimes n}$$

$$|\psi(x, \theta_Q)\rangle = U(\theta_Q)|\psi(x)\rangle$$

$$z_Q = \langle \psi(x, \theta_Q) | O | \psi(x, \theta_Q) \rangle$$

The combined learning of quantum and classical models raise some interesting questions:

- ▶ How do they interact with the learning? [7],[25]
- ▶ How do we know if the quantum layer is learning? (Using trend in gradient norm![3])
- ▶ If the quantum layer is learning does it mean it contributes positively to performance? (NO!)